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PPG, BVP, and Optical Heart Rate

Optical heart rate is also referred to as:

- PPG = Photoplethysmography
- BVP = Blood Volume Pulse

PPG sensors currently integrated in iMotions:

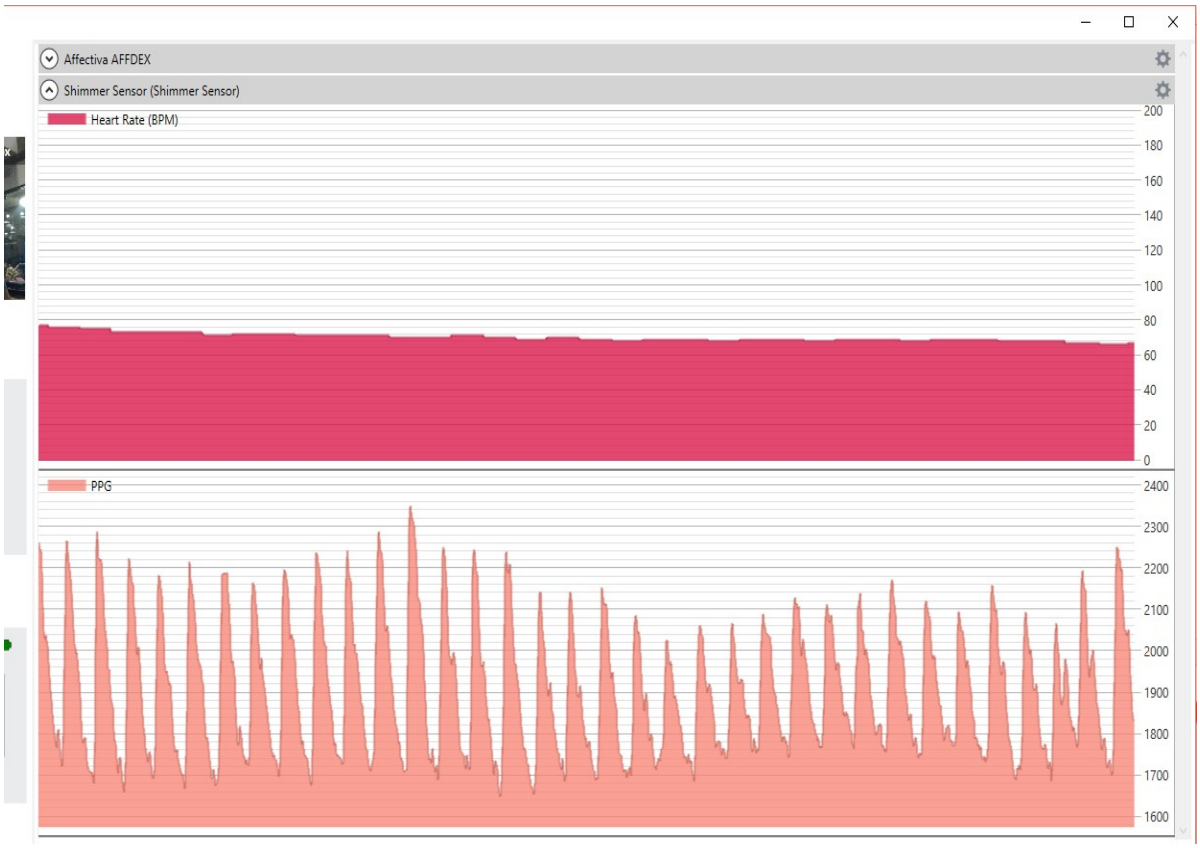
Click through the respective links for information regarding setup and configuration for each of the following devices:

- Shimmer3: This device comes with one PPG channel. Sensors can be attached to the finger (velcro) or earlobe (clip).
- Empatica E4. This device has one PPG channel. The sensor is monitoring BVP from the wrist.
- BIOPAC PPG100C. This BIOPAC module provides one PPG channel. Sensors can be placed on various locations.

Data output

In both the live viewer and replay views, you can see the PPG signal as well as a continuously calculated beats per minute value. When setting up a study with PPG, make sure you have clear peaks observed as shown in the second graph below before beginning the recording. Observe what can cause noise to these peaks and be aware of how those movements will affect your data.

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Please note Heart rate variability can be calculated from electroencephalographic (ECG) data (not with optical PPG data) using R Notebooks in iMotions. The supported ECG devices are Polar Belt H10, Shimmer ECG, Biopac ECG, and Bionomadix ECG. Read here to learn more about carrying out this analysis.

References

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